A nonminimizing limit of the uncorrected noisy game

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Abstract

The answer to C73-25, with its stated admissible class, is negative. We construct fixed smooth potential costs in three states and smooth, nonnegative repulsions tending locally uniformly to zero. Their exact uncorrected master equations have unique classical solutions. The actual selected population laws converge to a deterministic path which is not the population path of any global potential minimizer. The selection argument is an area identity along the actual noisy population.

Scope. The canonical statement permits a vanishing interior perturbation of the boundary repulsion, with no monotonicity or support restriction. The example uses that permission. It does not settle the more restricted question in which the penalty must be a decreasing cutoff supported only near zero. The original costs remain unchanged. The cost-corrected selection theorem in [3, Theorem 4.5, p. 3692] and [2] has different equations.

1 The equation and the conclusion

Write $\Delta_3 = \{m \geq 0 : \sum_i m_i = 1\}$ and $A_m = \operatorname{diag}(m) - mm^\top$. Set $T = 1$ and $\mathcal{F} = 0$. For $U \in \mathbb{R}^3$ put

$$a_{ij} = (U_i - U_j)_+, \quad H_i(U) = \tfrac{1}{2}\sum_{j\neq i} a_{ij}^2, \quad b_i^a = \sum_{j\neq i}(m_j a_{ji} - m_i a_{ij}).$$

Given a penalty ϕ, set

$$r_i^\phi = \phi(m_i) - m_i \sum_j \phi(m_j), \qquad Q_i = \sum_j \phi(m_j)(U_j - U_i), \qquad b = b^a + r^\phi.$$

The exact master equation used below is

$$\partial_t U_i + \varepsilon A_m : D^2 U_i + [b + 2\varepsilon(e_i - m)] \cdot DU_i - H_i(U) + Q_i = 0, \qquad U_i(1, m) = \partial_i \mathcal{G}(m). \tag{1}$$

All derivatives are tangent derivatives. The selected population has drift $b(m, U(t, m))$ and covariance $2\varepsilon A_m\, dt$, with the Brownian realization specified in C73-25. In particular, its noise is the original Kimura noise.

Theorem 1. *There are a fixed smooth ambient potential $\mathcal{G}$, a fixed $m_0 \in \operatorname{int}\Delta_3$, and an admissible penalty family ϕ_ε for which the actual solutions of (1) and their selected populations satisfy*

$$m^\varepsilon \longrightarrow \bar{m} \quad \text{in probability in } C([0,1], \Delta_3), \qquad \bar{m} \notin \mathcal{M}_{1,m_0}.$$

Thus their limiting law $Q = \delta_{\bar{m}}$ satisfies $Q(\mathcal{M}_{1,m_0}) = 0$.

More precisely, for one sufficiently small fixed $\delta > 0$, the construction has

$$\eta_\varepsilon = \frac{1}{\log(e + 1/\varepsilon)}, \qquad \theta_\varepsilon = \min\{\varepsilon^2, 1/100, 1/(16\sqrt{3})\}, \qquad \phi_\varepsilon(s) = \kappa_\varepsilon \chi(s/\theta_\varepsilon)^2 + \eta_\varepsilon \psi_\delta(s). \tag{2}$$

Here χ is smooth, nonincreasing, one on $[0,1]$ and zero on $[2,\infty)$; $\psi_\delta \geq 0$ is fixed and smooth. A sufficient height κ_ε is specified in Lemma 2. There is no restriction on the product $\kappa_\varepsilon\theta_\varepsilon$. In addition,

$$\mathbb{E}\int_0^1 \|r^{\phi_\varepsilon}(m_t^\varepsilon)\|_1\, dt \longrightarrow 0,$$

and the population laws are tight in the required path space.

Constants denoted C are independent of the small parameters until those parameters are fixed. A subscript δ permits dependence on the fixed construction, including its small radial scale. The order of choices is: fixed cutoffs, a small fixed ρ_0, the absorption constant K_0 determined by the uniform profile and forward Poincaré bounds, then a sufficiently small fixed δ and $R = \delta^2$, and finally $\eta = \eta_\varepsilon \to 0$ and $\varepsilon \to 0$.

2 Exact flat coordinates in one ordering chamber

On the chamber $U_1 > U_2 > U_3$ introduce

$$y_1 = \sqrt{2m_1}, \qquad y_2 = \sqrt{2m_1 + 4m_2}, \qquad P_1 = y_1\left(U_1 - \tfrac{U_2+U_3}{2}\right), \qquad P_2 = \tfrac{y_2}{2}(U_2 - U_3). \tag{3}$$

The inverse map is

$$m_1 = y_1^2/2, \quad m_2 = (y_2^2 - y_1^2)/4, \quad m_3 = 1 - (y_1^2 + y_2^2)/4.$$

Consequently $U \cdot dm = P \cdot dy$ and

$$\overline{H}(m, U) := \sum_i m_i H_i(U) = \tfrac{1}{2}|P|^2, \qquad U_1 > U_2 > U_3 \iff P_1/y_1 > P_2/y_2 > 0. \tag{4}$$

Indeed, the three positive differences are $P_1/y_1 - P_2/y_2$, $P_1/y_1 + P_2/y_2$, and $2P_2/y_2$. Inserting them into $\sum_i m_i H_i$ proves (4). Thus the original deterministic feedback dynamics with zero running cost, in this chamber and without repulsion, are $\dot{y} = -P$, $\dot{P} = 0$.

For an interior penalty $\phi = \eta\psi$, put $\Psi = \sum_i \psi(m_i)$ and define

$$\begin{aligned} R_1(y) &= \frac{\psi(m_1)}{y_1} - \frac{\Psi}{2} y_1, \\ R_2(y) &= \frac{2\psi(m_2) + \psi(m_1)}{y_2} - \frac{\Psi}{2} y_2, \\ D(y) &= \operatorname{diag}(\Psi + R_1/y_1, \Psi + R_2/y_2). \end{aligned} \tag{5}$$

The exact deterministic equations of the uncorrected game are

$$\dot{y} = -P + \eta R(y), \qquad \dot{P} = \eta D(y) P. \tag{6}$$

For example, with $J_1 = y_1$, $J_2 = y_2/2$ and $p_{v,k} = P_k/J_k$, the equations for component differences give $\dot{p}_{v,k} = c_k p_{v,k}^2/2 + \eta\Psi p_{v,k}$, where $c_1 = 2, c_2 = 1$. Also $\dot{J}_k = -c_k P_k/2 + \eta(R_k/y_k)J_k$. The quadratic terms cancel in $\dot{P}_k$ and give (6). There is no derivative of ψ in this costate equation.

3 A fixed nonnegative profile with centered rotation

Let

$$m_* = (1/5, 3/10, 1/2), \quad u = \sqrt{2/5}, \quad y_* = (u, 2u), \quad p = (1, 1), \quad P_* = \delta p, \quad y_0 = y_* + \delta p.$$

Set $m_0 = m(y_0)$, using the inverse map in Section 2. Take disjoint fixed smooth cutoffs ζ_1, ζ_2, equal to one near $1/5, 3/10$ respectively and supported in small interior intervals disjoint from $1/2$. For parameters h_1, h_2, d_1, d_2 set

$$h_a(s) = \sum_{j=1}^{2} [h_j + d_j(s - s_j)]\zeta_j(s), \qquad s_1 = 1/5, \quad s_2 = 3/10, \qquad \psi_{\delta,a}(s) = s + \delta h_a(s).$$

The baseline $\psi(s) = s$ has $R = 0$ and $D = I$ in (5). Linearity therefore gives

$$R_{\delta,a} = \delta R_h, \qquad D_{\delta,a} = I + \delta D_h, \qquad M_h = D_h - D R_h. \tag{7}$$

Here DR_h means the ordinary Jacobian with respect to y.

Fix the two slopes at

$$d_1 = -\frac{5}{7u}, \qquad d_2 = -\frac{1}{7u}.$$

Choose the two values by the following exact linear equations:

$$Z_\delta(a) := -\int_0^1 \left[R_{\delta,a}(y_* + s\delta p) + (1-s)D_{\delta,a}(y_* + s\delta p)\delta p\right] ds = 0. \tag{8}$$

The map Z_δ/δ is smooth at $\delta = 0$, where it equals $-R_h(y_*) - p/2$. Direct substitution in (5) gives

$$\frac{R_{h1}(y_*)}{u} = 2h_1 - \tfrac{1}{2}h_2, \qquad \frac{R_{h2}(y_*)}{2u} = \tfrac{1}{8}h_1 + \tfrac{3}{4}h_2.$$

The values-to-drift matrix has determinant $25u^2/8 \neq 0$. Thus, for all sufficiently small δ, inversion of a smooth two-by-two matrix gives unique smooth values $h_1(\delta), h_2(\delta)$ satisfying (8), with limits

$$h_1(0) = -\frac{8}{25u}, \qquad h_2(0) = -\frac{7}{25u}. \tag{9}$$

The functions $h_{a(\delta)}$ have uniform smooth bounds. Outside their fixed interior supports $\psi_\delta(s) = s$; on the supports s is bounded below. Hence $\psi_\delta = s + \delta h_{a(\delta)} \geq 0$ on $[0,1]$ after reducing δ. This fixes one smooth scalar profile.

We record the matrix calculation responsible for the selection. At y_* let $K_h = 15h_1/4 - 5h_2/2$. Differentiating (5) yields

$$\begin{aligned}(M_h)_{11} - (M_h)_{22} &= K_h - 4d_1/5 + d_2/2,\\ (M_h)_{12} + (M_h)_{21} &= -d_1/10 + d_2/2,\\ (M_h)_{12} &= d_2/5.\end{aligned}$$

At the limiting values (9) these give

$$M_h(y_*) = \frac{1}{u}\begin{pmatrix} -113/70 & -1/35 \\ 1/35 & -113/70 \end{pmatrix} = \mu_0 I + \omega_0 J, \qquad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \omega_0 = -\frac{1}{35u} < 0. \tag{10}$$

For our values at positive δ, the error in this matrix is $O(\delta)$. Freeze at y_* the matrices

$$A = DR_\delta(y_*), \quad D = D_\delta(y_*), \quad B = D - A, \quad C = J\left(B - \tfrac{1}{2}\operatorname{tr}(B)I\right), \quad k = \operatorname{tr} D - \tfrac{1}{2}\operatorname{tr} B. \tag{11}$$

Every traceless two-by-two matrix becomes symmetric after left multiplication by J. By (7)–(10),

$$A = O(\delta), \quad D = I + O(\delta), \quad k = 1 + O(\delta), \quad c_0\delta I \leq C \leq c_1\delta I, \quad A^\top + D - kI = O(\delta) \tag{12}$$

for positive fixed constants c_0, c_1. In fact $C = \delta|\omega_0|I + O(\delta^2)$.

4 Fixed smooth costs and a deterministic reference

Let

$$c = (3\delta/(2u), \delta/u, 0), \qquad \mathcal{G}_{\mathrm{b}}(m) = c \cdot m.$$

Its flat gradient at y_* is P_*. Choose once and for all a smooth bounded function q on $[0,\infty)$, equal to -2 on $[0,1]$ and zero on $[4,\infty)$. With $R = \delta^2$ set

$$H_R(r) = \int_0^r tq(t^2/R^2)\, dt. \tag{13}$$

Then $H_R(r) = -r^2$ for $r \leq R$, its gradient vanishes for $r \geq 2R$, and

$$|\nabla H_R| \leq CR, \qquad \|D^2 H_R\| + |H_R'(r)/r| \leq C, \qquad \|D^3 H_R\| \leq C/R. \tag{14}$$

The value at $r = 0$ in the quotient is interpreted by continuity. The composition is a smooth radial function of y.

On a flat ball of fixed radius ρ_0 about y_* define the desired local terminal potential

$$\widetilde{\mathcal{G}}(y) = \mathcal{G}_{\mathrm{b}}(m_*) + P_* \cdot (y - y_*) + H_R(|y - y_*|). \tag{15}$$

Multiply $\widetilde{\mathcal{G}} - \mathcal{G}_{\mathrm{b}}(m(y))$ by a smooth cutoff equal to one on radius $\rho_0/2$ and zero outside radius ρ_0, and add $\mathcal{G}_{\mathrm{b}}$. Choose the ball compactly inside the simplex chart. This defines a smooth potential on the whole simplex. Its degree-one homogeneous extension $\mathcal{G}(m) = S\mathcal{G}(m/S)$, $S = \sum_i m_i > 0$, defines a smooth ambient potential on a neighborhood of the simplex. Since the linear flat term and $\mathcal{G}_{\mathrm{b}}$ agree to first order at y_*, this extension has

$$\max_i \|g_i - c_i\|_\infty \le C(\delta\rho_0 + R), \qquad K := \max_i \|g_i\|_\infty \le C\delta. \tag{16}$$

All these costs are fixed before ε varies.

For small $\eta \ge 0$, solve (6) backward with terminal position z near y_* and momentum $\nabla\widetilde{\mathcal{G}}(z)$. At $\eta = 0, z = y_*$ this is $\bar{y}^0(t) = y_* + (1-t)P_*$, $\bar{P}^0(t) = P_*$. The terminal-to-initial position derivative is $I + D^2\widetilde{\mathcal{G}}(y_*) = -I$, hence is invertible. Differentiating the backward ODE with respect to η at this reference gives exactly the initial-position derivative Z_δ in (8): in backward time s the momentum derivative is $-\int_0^s D_\delta(y_* + r\delta p)\delta p\, dr$, and its position derivative is its integral minus $\int_0^s R_\delta(y_* + r\delta p)\, dr$. Therefore (8) and the implicit function theorem give an exact deterministic reference $(\bar{y}^\eta, \bar{P}^\eta)$ with

$$\bar{y}^\eta(0) = y_0, \qquad \bar{y}^\eta(1) = y_* + s_\eta, \quad |s_\eta| \le C_\delta\eta^2, \qquad |\bar{y}^\eta - y_*| + |\bar{P}^\eta| \le C\delta. \tag{17}$$

It converges uniformly to $(\bar{y}^0, P_*)$. This reference is used for centering.

5 Admissibility and estimates independent of layer height

Lemma 2. *Let $q_0 \ge 0$ be any fixed smooth interior addition with $\|q_0\|_\infty \le Q$. For fixed $0 < \varepsilon < 1/2$ and $0 < \theta < 1/(4\sqrt{3})$, the penalty $\phi(s) = \kappa\chi(s/\theta)^2 + q_0(s)$ is admissible for (1) if*

$$\kappa \ge \max\{\kappa_H(\sqrt{2\varepsilon}, 2K + Q + 2\varepsilon), 1 + \varepsilon\}. \tag{18}$$

Here $\kappa_H(\sigma, B_0)$ is the sufficient threshold of [1, Theorem 2.10]. It depends on noise and the bound B_0 for the nonnegative inward coefficient, and not on θ or the large outflow bound. The solution is unique in the canonical classical class and satisfies $|U_i| \le K$. The same statement holds on any fixed finite horizon, with $K = \|g\|_\infty$ since $\mathcal{F} = 0$.

Proof. We give the reduction to the cited linear theory, since the printed master theorem [1, Theorem 3.4] assumes a decreasing cutoff and cannot directly be applied to our total penalty. Put $a_M(r) = \min(M, r_+)$, $M = 2K$, and use $H_M^{-,i}(v) = -\sum_{j\ne i}\int_0^{(v_i - v_j)_+} \min(M, r)\, dr$. With $\phi_0 = \kappa\chi(s/\theta)^2$, the generic fixed-point system of [1, Section 4.3.2, (4.16)–(4.18)] has component drift $\phi_0(m_j) + \beta_j^i + m_j\beta_j^\circ$, where we insert

$$\beta_j^i = q_0(m_j) + \sum_\ell m_\ell a_M(v_\ell - v_j) + 2\varepsilon\delta_{ij},$$
$$\beta_j^\circ = -\sum_\ell [\phi(m_\ell) + a_M(v_j - v_\ell)] - 2\varepsilon.$$

These coefficients sum to zero in the complete drift and give precisely the clipped population drift plus $2\varepsilon(e_i - m)$. Crucially, $0 \le \beta_j^i \le Q + M + 2\varepsilon$; the large κ appears in the outflow coefficient only. If P_Γ clips to $[-\Gamma, \Gamma]$, use the bounded source

$$P_\Gamma\left[H_M^{-,i}(v) + \sum_j \phi(m_j)P_\Gamma(v_j - v_i)\right].$$

For fixed parameters all the coefficient regularity assumptions of that construction hold. Its frozen linear solvability [1, Lemma 4.4], the Hölder estimate [1, Theorem 2.10], and the Schauder argument in its Section 4.3.2 yield a Wright–Fisher classical solution of this clipped system, with bounded first derivatives. Their constants may depend on the fixed outflow and source sizes; the repulsion threshold uses the displayed inward bound. This is why (18) is noncircular.

At a maximum over all components, $H_M^{-,i} \le 0$ and the switching source is nonpositive; at a minimum $H_M^{-,i} = 0$ and the switching source is nonnegative. The maximum principle is justified at the degenerate boundary using $B_{\mathrm{ent}} = \log 3 + \sum_j m_j \log m_j$. For any positive α, a maximum of $U_i - \alpha B_{\mathrm{ent}}$ cannot lie on the boundary: an inward displacement changes U_i by $O(h)$ and $-\alpha B_{\mathrm{ent}}$ by a positive multiple of $-\alpha h \log h + O(h)$. The component generator applied to B_{ent} is bounded above for fixed parameters, since $\varepsilon A_m : D^2 B_{\mathrm{ent}} = 2\varepsilon$ and the other entropy drifts have finite upper bounds. Subtract also this bound plus one times $\alpha(1-t)$, and apply the interior maximum principle in reverse time. Letting $\alpha \downarrow 0$ gives $|U_i| \le K$. Thus $M = 2K$ removes the control and Hamiltonian clips. After κ is fixed choose $\Gamma > \max\{2K, 4K^2 + 6K\|\phi\|_\infty\}$; both source clips are then inactive. This proves existence for the exact unchanged costs. If $K = 0$, $U = 0$ works, or one can use any positive control clip.

For bounded feedback rates $a_{ij} \le M_1$, $b_j \ge \phi(m_j) - C_\varepsilon m_j$. On $m_j \le \theta$, the generator of the population, or of any component diffusion with added drift $2\varepsilon(e_i - m)$, applied to $-\log m_j$ is at most $-(\kappa - \varepsilon)/m_j + C_\varepsilon + 2\varepsilon$. Off that layer it is bounded above. Stopped logarithmic barriers exclude hitting any face. Interior local Lipschitz coefficients give a unique strong solution extending throughout $[0, 1]$.

Finally, if U, V are canonical classical solutions, their difference w_i satisfies a linear equation with the component generator based on U and source

$$\sum_j \phi(m_j)(w_j - w_i) - H_i(U) + H_i(V) + [b(m, U) - b(m, V)] \cdot DV_i.$$

For fixed ε this source is bounded by $C_\varepsilon \max_j \|w_j\|_\infty$, including the drift mismatch term because DV is bounded. Itô's formula along the interior component diffusion, followed by removal of compact stopping times, gives backward Gronwall with zero terminal data. Hence $w = 0$. □

Apply the lemma in (2) with $q_0 = \eta_\varepsilon \psi_\delta$ and $Q = \|\psi_\delta\|_\infty$, taking κ_ε at least the right side of (18) and $1/\varepsilon$. The noise convention in [1] is generator $\sigma^2 A_m/2$, so $\sigma = \sqrt{2\varepsilon}$ here. The cutoff is supported in $[0, 2\theta_\varepsilon]$; hence the family is smooth, nonnegative and locally uniformly vanishing on $(0, 1]$. Only small ε matters. If an index for every $\varepsilon > 0$ is wanted, put $a = \max(1, 4\varepsilon)$ and apply the lemma to $V(s, m) = U(s/a, m)/a$ with horizon a, noise ε/a, terminal data g/a and penalty ϕ/a; rescaling back supplies the remaining indices without changing g.

Two useful uniform estimates also explain why an arbitrarily high layer causes no hidden impulse. With $M = 2K$, put $\mu = \min_i m_{0i}$, $C_* = 2(M + 1)$ and $\ell = (\mu/2)e^{-C_*}$. Let $\alpha_\varepsilon = \sup_{s \in [\ell, 1]} \phi_\varepsilon(s) \to 0$. Stop the actual population on its first coordinate reaching ℓ. Before this time $b_i + C_* m_i \ge 0$, for small ε. The integrating-factor martingale has expected squared supremum at most $2\varepsilon e^{2C_*}$, since $d\langle M_i^m \rangle \le \varepsilon\, dt/2$. A coordinate exit needs a martingale drop of at least $\mu/2$. Thus

$$\mathbb{P}(\text{exit before } 1) \le 24\varepsilon e^{2C_*}/\mu^2. \tag{19}$$

On nonexit paths the integrated repulsion is bounded by $3\alpha_\varepsilon$. The controlled drift is uniformly bounded and the population martingale tends uniformly to zero in L^2. These facts prove tightness and repulsion removal in probability.

For expected absolute removal, set $F_{\mathrm{ent}} = \sum_i m_i \log m_i$ and $k_\ell = \log(1/(3\ell)) > 0$. Its repulsion drift is

$$r^\phi \cdot DF_{\mathrm{ent}} \le -k_\ell \sum_{i : m_i < \ell} \phi(m_i) + 3\alpha_\varepsilon \log 3.$$

Its controlled drift is at most $2M \log 3$, its diffusion drift is 2ε, and its martingale bracket is bounded by $24\varepsilon/e^2\, dt$. Stop first inside the simplex and then pass to the limit; the negative term is integrable by Fatou and the other terms have the displayed bounds. On the exit event, apply the resulting entropy

inequality from the exit time to 1 conditionally on that event. The entropy oscillation is at most $\log 3$. Together with (19), this gives

$$\mathbb{E}\int_0^1 \sum_i \phi_\varepsilon(m_i^\varepsilon)\,dt \le 3\alpha_\varepsilon + C_\ell\varepsilon \longrightarrow 0. \tag{20}$$

Since $\|r^\phi\|_1 \le 2\sum_i \phi(m_i)$, this proves the claimed expected removal without a height-times-width assumption.

6 The actual feedback remains in the protected chamber

We now reduce ρ_0 and then δ as necessary. The value estimate and (16) give $|U_i| \le C\delta$ globally, so all actual rates are $O(\delta)$. For a component i, use the auxiliary diffusion with generator

$$\varepsilon A_m : D^2 + [b(m,U) + 2\varepsilon(e_i - m)]\cdot D.$$

Start in the flat ball of radius $\rho_0/2$ and stop on radius ρ_0. On this protected set the boundary cutoff is absent, $\phi = \eta\psi_\delta$, and the drift is $O(\delta+\eta+\varepsilon)$. Choose δ small compared with ρ_0 and then $\eta, \varepsilon \le \delta$. A deterministic drift displacement is less than $\rho_0/4$, so Doob's inequality bounds the exit probability by $C\varepsilon/\rho_0^2$. Along this auxiliary diffusion (1) gives $dU_i = (H_i - Q_i)\,dt +$ martingale. Before exit, $|H_i| \le C\delta^2$ and $|Q_i| \le C\eta\delta$. Using (16) at the terminal time, and the global value bound at a stopped exit, proves uniformly on the inner ball and in time

$$|U_i(t,m) - c_i| \le C(\delta\rho_0 + R + \delta^2 + \eta\delta + \delta\varepsilon/\rho_0^2). \tag{21}$$

The gaps of c are positive fixed multiples of δ. Choose ρ_0, then δ with $R = \delta^2$, and finally η, ε so that the right side is less than one quarter of the smallest gap. This proves $U_1 > U_2 > U_3$ for the actual global master field throughout the inner ball.

Let L be a fixed sufficiently large constant and stop the actual selected flat population at

$$\tau = 1 \wedge \inf\{t : |y_t - y_0| \ge L\delta\}.$$

Reduce δ so that this ball and the reference (17) are inside radius $\rho_0/4$ about y_*. The actual flat population drift is $O(\delta)$ and its bracket is $O(\varepsilon)$. Taking L greater than twice the deterministic drift bound gives

$$\mathbb{P}(\tau < 1) \le C\varepsilon/\delta^2. \tag{22}$$

Up to this stop the actual and reference positions are $O(\delta)$ from y_*, their momenta are $O(\delta)$, and the exact flat and radial formulas hold.

7 Original weighted energy and exact finite-noise errors

Along the actual selected population write $z_i = (e_i - m)\cdot DU_i$ and $\Gamma_i = DU_i^\top A_m DU_i$. For $V = \sum_i m_i U_i^2$, the generator including time gives

$$(\partial_t + \varepsilon A_m : D^2 + b\cdot D)V = 2\varepsilon\sum_i m_i\Gamma_i - \sum_{i,j\ne i} m_i U_j a_{ij}^2 + \mathcal{R}, \quad \mathcal{R} = \sum_j \phi(m_j)\sum_i m_i(U_j - U_i)^2 \ge 0. \tag{23}$$

Indeed, the cross term $4\varepsilon U_i\Gamma_{A_m}(m_i,U_i) = 4\varepsilon m_i U_i z_i$ cancels the weighted-player contribution $-4\varepsilon m_i U_i z_i$. On an active edge, $a_{ij}(U_j^2 - U_i^2) + U_i a_{ij}^2 = -U_j a_{ij}^2$. The repulsion terms combine as $\sum_i r_i U_i^2 - 2\sum_i m_i U_i Q_i = \mathcal{R}$. These identities prove (23).

Stop on $\min_i m_i \ge 1/n$, apply Itô's formula and take expectations. The right side after rearrangement is bounded by $K^2 + 2K(2K)^2$. The nonnegative left integrands pass by monotone convergence, and the bounded remaining terms pass by bounded convergence, because each continuous interior path stays in some compact interior set. Hence

$$2\varepsilon\mathbb{E}\int_0^1 \sum_i m_i\Gamma_i\,dt + \mathbb{E}\int_0^1 \mathcal{R}\,dt \le C_E, \qquad C_E = K^2 + 8K^3 = O(\delta^2). \tag{24}$$

In particular, on the actual stopped protected set, $\varepsilon\mathbb{E}\int_0^\tau |D_{\tan}U|^2\,dt \le C_\delta$ and $\varepsilon\mathbb{E}\int_0^\tau |D_{\tan}U|\,dt \le C_\delta\sqrt{\varepsilon}$. Let B_v have rows $(1,-1/2,-1/2)$ and $(0,1,-1)$, so that $p_v = B_vU$. Let M^m be the population martingale, $K_y = Dy\,A_m(Dy)^\top$, and put

$$\ell_k = \sum_i (B_v)_{ki} z_i, \qquad t_k = \sum_i (B_v)_{ki} DU_i^\top A_m D_m y_k.$$

On $[0,\tau]$, the exact equations are

$$\begin{aligned} dy_k &= [-P_k + \eta R_k - \varepsilon K_{y,kk}/y_k]\,dt + dM_k^y,\\ dP_k &= [\eta D_{kk}(y)P_k + \varepsilon e_k]\,dt + dN_k^P,\\ e_k &= -K_{y,kk}P_k/y_k^2 - 2J_k\ell_k + c_k t_k,\\ dM_k^y &= D_m y_k \cdot dM^m,\\ dN_k^P &= J_k \sum_i (B_v)_{ki} DU_i \cdot dM^m + (P_k/y_k) D_m y_k \cdot dM^m. \end{aligned} \tag{25}$$

To check them, the original equation along the population is $dU_i = (H_i - Q_i - 2\varepsilon z_i)\,dt + DU_i \cdot dM^m$. Thus $dp_{v,k} = (c_k p_{v,k}^2/2 + \eta\Psi p_{v,k} - 2\varepsilon\ell_k)\,dt + \sum_i (B_v)_{ki} DU_i \cdot dM^m$. In $d(J_k p_{v,k})$ the quadratic drift cancels as before, the coordinate Itô drift gives the first term of e_k, and the product bracket gives $\varepsilon c_k t_k$. This proves (25). Bounded geometry on the protected set and (24) imply

$$\begin{aligned} \mathrm{TV}(E_X) &\le C_\delta\varepsilon, \qquad & \mathbb{E}\,\mathrm{TV}(E_p) &\le C_\delta\sqrt{\varepsilon},\\ \mathbb{E}\,\mathrm{tr}\langle M^y\rangle_\tau &\le C_\delta\varepsilon, \qquad & \mathbb{E}\,\mathrm{tr}\langle N^P\rangle_\tau &\le C_\delta, \end{aligned} \tag{26}$$

where $dE_{X,k} = -\varepsilon K_{y,kk}/y_k\,dt$ and $dE_{p,k} = \varepsilon e_k\,dt$. In particular the costate martingale is bounded in energy. Cauchy–Schwarz for covariations gives

$$\mathbb{E}\,\mathrm{TV}\langle M_i^y, N_j^P\rangle_\tau \le C_\delta\sqrt{\varepsilon}. \tag{27}$$

8 An area estimate for the actual centered processes

Center the actual variables at the reference with $\eta = \eta_\varepsilon$: $X = y - \bar{y}^\eta$, $p_c = P - \bar{P}^\eta$, stopping both at the same τ. In this section abbreviate p_c to v. Equations (17), (7) and Taylor's theorem give the exact decomposition

$$\begin{aligned} dX &= [-v + \eta AX + \eta r_X]\,dt + dE_X + dM^y,\\ dv &= [\eta Dv + \eta r_v]\,dt + dE_p + dN^P,\\ |r_X| &\le C\delta^2|X|, \qquad |r_v| \le C\delta^2(|X| + |v|), \qquad X_0 = 0. \end{aligned} \tag{28}$$

For the second remainder, explicitly write $[D(y) - D(y_*)]v + [D(y) - D(\bar{y})]\bar{P}$. Both derivatives of the nonconstant part of D are $O(\delta)$; the positions and $\bar{P}$ are $O(\delta)$. The first remainder follows similarly from $DR(y) - DR(y_*) = O(\delta|y - y_*|)$.

On the good event $G = \{\tau = 1\}$, the terminal radial formula gives

$$v_1 = h_1 X_1 + e_1, \qquad |h_1| \le H, \qquad |e_1| \le C_\delta\eta^2|X_1|. \tag{29}$$

Indeed, $v_1 = \nabla H_R(X_1 + s_\eta) - \nabla H_R(s_\eta)$. Take $h_1 = H_R'(|X_1|)/|X_1|$. The mixed difference from $\nabla H_R(X_1)$ is bounded by $\|D^3 H_R\|\,|s_\eta|\,|X_1|$. Use (14) and (17); H is uniform in small δ, whereas C_δ is allowed to depend on R.

First, the forward equation in (28), Gronwall, Doob and Cauchy–Schwarz give

$$\mathbb{E}\sup_{t\le 1}|X_{t\wedge\tau}|^2 \le C\mathbb{E}\int_0^\tau |v_t|^2\,dt + C_\delta\varepsilon. \tag{30}$$

For example, its square root follows from the pathwise bound $\sup|X| \le 2(\int_0^\tau |v|\,dt + \mathrm{TV}(E_X) + \sup|M^y|)$, valid once $\eta\|A\| + C\eta\delta^2$ is small. It also bounds $\mathbb{E}\int_0^\tau w|X|^2\,dt$ by $C\mathbb{E}\int_0^\tau w|v|^2\,dt + C_\delta\varepsilon$ for $w(t) = e^{-\eta k t}$, since $1/2 \le w \le 1$ for small η.

Choose K_0 large, put $\lambda = K_0\delta^2$ and set

$$N = J + \eta\lambda I, \quad Q = X^\top N v, \quad E = \tfrac{1}{2}X^\top C X, \quad L_c = A^\top + D - kI.$$

The two-by-two identity $A^\top J + JD = C + kJ$, together with Itô's formula, gives

$$\begin{aligned} dQ - \eta k Q\,dt &= [\eta X^\top C v - \eta\lambda|v|^2 + \eta^2\lambda X^\top L_c v]\,dt \\ &\quad + \eta[r_X^\top N v + X^\top N r_v]\,dt + d\mathcal{V}_Q + d\mathcal{M}_Q, \\ dE &= [-X^\top C v + \eta X^\top C A X + \eta X^\top C r_X]\,dt + d\mathcal{V}_E + d\mathcal{M}_E, \end{aligned}$$

where the finite-variation noise terms are

$$\begin{aligned} d\mathcal{V}_Q &= (dE_X)^\top N v + X^\top N\,dE_p + \sum_{ij} N_{ij}\,d\langle M_i^y, N_j^P\rangle, \\ d\mathcal{V}_E &= X^\top C\,dE_X + \tfrac{1}{2}\sum_{ij} C_{ij}\,d\langle M_i^y, M_j^y\rangle. \end{aligned}$$

All martingale terms have mean zero after stopping, by bounded X, v and (26). Integrating with factor w, substituting the energy equation and using $X_0 = 0$ gives the exact identity

$$\begin{aligned} &\mathbb{E}[w(\tau)(Q_\tau + \eta E_\tau)] + \eta\lambda\mathbb{E}\int_0^\tau w|v|^2\,dt \\ &\quad + \eta^2\mathbb{E}\int_0^\tau w X^\top[\tfrac{k}{2}C - \mathrm{Sym}(CA)]X\,dt \\ &= \eta^2\lambda\mathbb{E}\int_0^\tau w X^\top L_c v\,dt + \eta\mathbb{E}\int_0^\tau w[r_X^\top N v + X^\top N r_v]\,dt \\ &\quad + \eta^2\mathbb{E}\int_0^\tau w X^\top C r_X\,dt + \mathbb{E}\int_0^\tau w\,d\mathcal{V}_Q + \eta\mathbb{E}\int_0^\tau w\,d\mathcal{V}_E. \end{aligned} \tag{31}$$

By (26)–(27), the last two terms in absolute value are at most $C_\delta\sqrt{\varepsilon}$. By (12), after reducing δ,

$$\tfrac{k}{2}C - \mathrm{Sym}(CA) \geq c_2\delta I.$$

The first-order remainder on the right of (31) is bounded by $C\eta\delta^2\mathbb{E}\int w(|X|^2 + |X||v|)\,dt$. Use (30) and $2|X||v| \leq |X|^2 + |v|^2$. Choosing K_0 sufficiently large absorbs it into one quarter of $\eta\lambda\mathbb{E}\int w|v|^2\,dt$, with error $C_\delta\varepsilon$. The $L_c = O(\delta)$ term is absorbed into another quarter by the same bound and small η. The remaining deterministic term is at most $C\eta^2\delta^3\mathbb{E}\int w|X|^2\,dt$ and is absorbed by the positive bulk matrix for small δ.

On G, (29) yields

$$Q_1 + \eta E_1 \geq \eta[c_0\delta/2 - \lambda H - C_\delta\eta - C_\delta\lambda\eta^2]|X_1|^2 \geq c_3\eta\delta|X_1|^2.$$

Here choose δ so that $\lambda H \leq c_0\delta/8$, then η small for that fixed δ. On exit, $|X| + |v| \leq C\delta$; the negative endpoint contribution has expectation at most $C\delta^2\mathbb{P}(\tau < 1) \leq C\varepsilon$ by (22). Consequently (31) proves

$$\eta\delta\mathbb{E}[\mathbf{1}_G|X_1|^2] + \eta\delta^2\mathbb{E}\int_0^\tau |v|^2\,dt \leq C_\delta\sqrt{\varepsilon}. \tag{32}$$

Since $\sqrt{\varepsilon}/\eta_\varepsilon \to 0$, (30), (22) and (32) imply $\sup_t |y_t - \bar{y}_t^{\eta_\varepsilon}| \to 0$ in probability. The reference convergence in (17) proves the path convergence in Theorem 1.

9 The limiting population path is not minimizing

The limit is the original central straight path $\bar{y}(t) = y_* + (1-t)P_*$, with $\bar{m} = m(\bar{y})$. Its feedback stays in the strict chamber: at y_*, $P_{*1}/y_{*1} = \delta/u > \delta/(2u) = P_{*2}/y_{*2} > 0$, and the inequalities persist along this short path.

It is necessary to exclude every control producing this population path, not just its equilibrium feedback. For any nonnegative rates a and cotangent U, completing the square edge by edge gives

$$L(m,a)+U\cdot b^a(m,a)+\overline{H}(m,U)\geq 0, \qquad L(m,a)=\tfrac{1}{2}\sum_{i,j\neq i} m_i a_{ij}^2. \tag{33}$$

Indeed each edge contributes $m_i[\frac{1}{2}a_{ij}^2-a_{ij}(U_i-U_j)+\frac{1}{2}(U_i-U_j)_+^2]\geq 0$. Along the central path choose the cotangent corresponding to P_* in (3). If any rates generate that same path, then $U\cdot\dot{m}=P_*\cdot\dot{\bar{y}}=-|P_*|^2$ and $\overline{H}=|P_*|^2/2$. Thus (33) gives $L\geq|P_*|^2/2$ almost everywhere. The central feedback attains this lower bound.

Take any fixed $w\in\mathbb{R}^2$ with $|w|=R/2$ and use the straight competitor from y_0 to y_*+w with constant momentum P_*-w. For sufficiently small fixed δ, $R=\delta^2$ ensures that this entire path lies in the protected strict chamber, so its original rates are nonnegative and its kinetic action is $|P_*-w|^2/2$. Its terminal potential is, by (15) and (13), $\mathcal{G}_{\mathrm{b}}(m_*)+P_*\cdot w-|w|^2$. Its total original action minus the minimum action among controls generating the central population path is therefore

$$\tfrac{1}{2}|P_*-w|^2+P_*\cdot w-|w|^2-\tfrac{1}{2}|P_*|^2=-\tfrac{1}{2}|w|^2<0.$$

This is a fixed positive action gap, independent of ε and η. No control producing $\bar{m}$ can be a global minimizer. Together with the actual path convergence, this proves Theorem 1. □

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

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